

Unit-1

Project Schedule

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- Project Scheduling

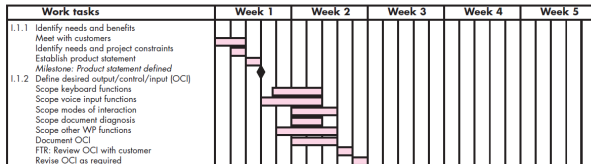
Introduction

- In software project schedule, a macroscopic schedule is developed.
- Each entry on the macroscopic schedule is refined into a detailed schedule.
- Specific software actions and tasks are identified and scheduled.

Project Scheduling

- Software project scheduling distributes estimated effort across the planned project duration by allocating the effort to specific tasks
- During early stages of project planning, a macroscopic schedule is developed identifying all major process framework activities and the product functions to which they apply.
- A time-line chart, also called a Gantt chart

Figure: Time-line chart



Scheduling(Contd..)

Software Tools

- Program evaluation and review technique (PERT) and the critical path method (CPM) are two project scheduling methods that can be applied to software development. Tools produce project tables.
- Majority of software project scheduling tools produce project tables a tabular listing of all project tasks, their planned and actual start and end dates, and a variety of related information

Figure: Project table

Work tasks	Planned start	Actual start	Planned complete	Actual complete	Assigned person	Effort allocated	Notes
1.1.1 Identify needs and benefits							
Meet with customers	wk1, d1	wk1, d1	wk1, d2	wk1, d2	BLS	2 p-d	Scoping will require more effort/time
Identify needs and project constraints	wk1, d2	wk1, d2	wk1, d2	wk1, d2	JPP	1 p-d	
Establish product statement	wk1, d3	wk1, d3	wk1, d3	wk1, d3	BLS/JPP	1 p-d	
Milestone: Product statement defined	wk1, d3	wk1, d3	wk1, d3	wk1, d3			
1.1.2 Define desired output/control/input (OCI)							
Scope keyboard functions	wk1, d4	wk1, d4	wk2, d2		BLS	1.5 p-d	
Scope online input functions	wk1, d3	wk1, d3	wk2, d2		IPP	2 p-d	

Basic principles of Software project scheduling:

- Compartmentalization, Interdependency, Time allocation, Effort validation, Defined responsibilities, Defined outcomes, Defined milestones

Scheduling(Contd..)

Case Study-Data provided in AGMS project

- CPM Estimation for the various activity & Flow of the activity in project

List of Activities for the Airport Gate Management System (AGMS) Acquisition Project

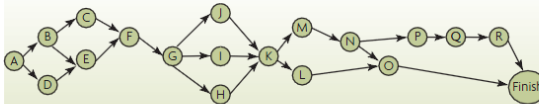
Activity	Description	Immediate Predecessors	Estimated Time (weeks)
A	Set up the project acquisition team	—	2
B	Write down the software requirements	A	2
C	Develop a contractor evaluation grid that will be used to evaluate proposals	B	1
D	Identify and select potential contractors	A	1
E	Develop and send out a request for proposal to potential contractors	B, D	4
F	Audit candidate contractors, select one contractor, negotiate and sign an agreement contract with the selected contractor	C, E	2
G	Prepare the definition of functional specifications	F	5
H	Develop a software testing plan	G	2
I	Software customization phase I	G	12
J	Purchase and install the hardware	G	2
K	Test the first release	H, I, J	1
L	Develop a training plan for key users	K	1
O	Train key users	L, N	2
M	Software customization phase II	K	6
N	Test the second release	M	1
P	Software customization phase III	N	3
Q	Test the final release	P	2
R	Software deployment and project sign-off	Q	4

Scheduling(Contd..)

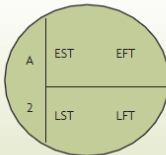
Case Study-Data provided in AGMS project

- CPM or Network diagram, total time to complete the project
- Early Start Time(EST) and Early Finish Time(EFT)
- Latest Start Time(LST) and Latest Finish Time (LFT)

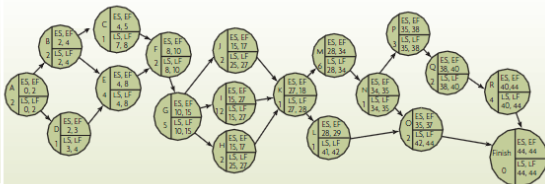
Network Representation for the AGMS Acquisition Project



Nodes Notation



LST and LFT Computation for the AGMS Acquisition Project



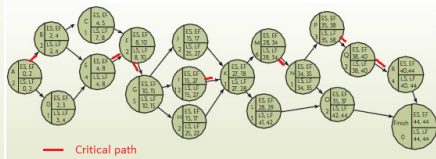
Scheduling(Contd..)

Case Study-Data provided in AGMS project

● Identification of Critical activity-CPM

Summary of Activities' Start, Finish, and Slack Times

Activity	EST	EFT	LST	LFT	Slack	Critical Activity
A	0	2	0	2	0	Yes
B	2	4	2	4	0	Yes
C	4	5	7	8	3	
D	2	3	3	4	1	
E	4	8	4	8	0	Yes
F	8	10	8	10	0	Yes
G	10	15	10	15	0	Yes
H	15	17	25	27	10	
I	15	27	15	27	0	Yes
J	15	17	25	27	10	
K	27	28	27	28	0	Yes
L	28	29	41	42	13	
M	28	34	28	34	0	Yes
N	34	35	34	35	0	Yes
O	35	37	42	44	7	
P	35	38	35	38	0	Yes
Q	38	40	38	40	0	Yes
R	40	44	40	44	0	Yes
Finish	44	44	44	44	0	Yes



Scheduling(Contd..)

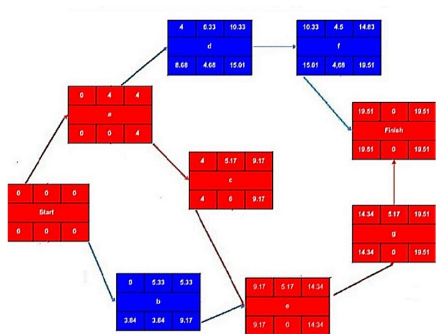
PERT

● Program Evaluation Review Technique method

Activity	Predecessor	Time estimates			Expected time
		Opt. (o)	Normal (m)	Pess. (p)	
A	—	2	4	6	4.00
B	—	3	5	9	5.33
C	A	4	5	7	5.17
D	A	4	6	10	6.33
E	B, C	4	5	7	5.17
F	D	3	4	8	4.50
G	E	3	5	8	5.17

Early Start	Duration	Early Finish
Task Name		
Late Start	Slack	Late Finish

$$\text{Expected time} = \frac{0+4m+p}{6}$$



Note the critical path is in red.

Scheduling(Contd..)

CPM

- Critical Path Method (CPM) is a project management tool which is suitable for projects that have predictable activities.
- CPM uses a single estimate for the time that a project can be completed .
- CPM is a deterministic project management tool
- CPM allows project management planners to determine which aspect of the project to sacrifice when a trade-off is needed in order to complete the project.

PERT

- The Program Evaluation and Review Technique (PERT) is a project management technique or tool which is suitable for projects that have unpredictable activities.
- PERT uses three estimates (optimistic, normal, pessimistic) for the time that it can be completed.
- PERT is a probabilistic project management tool.
- PERT does not allow project management planners to determine which aspect of the project to sacrifice when a trade-off is needed in order to complete the project.

- Why people are important in software project ?
- Impossible region in project schedule.
- To draw time line chart and task network diagram.
- Identify critical activity in schedule.

Questions

- What is the duration for this project completion ?
- What is the critical path ?
- How to calculate execution time in PERT estimation?
- Difference between PERT and CPM.
- Draw the time line chart.
- Complete the project table with the available information.

Figure: Sample project data

						4/1	4/8	4/15	4/22	4/29	5/6	5/13	5/20	5/27	6/3
Task #	TaskName	Duration	Start	Finish	Pred.										
A	Establish increments	3	4/1		None										
B	Analyze Inc One	3			A										
C	Design Inc One	8			B										
D	Code Inc One	7			C										
E	Test Inc One	10			D										
F	Install Inc One	5			E										
G	Analyze Inc Two	7			A, B										
H	Design Inc Two	5			G										
I	Code Inc Two	4			H										
J	Test Inc Two	6			E, I										

- During project scheduling, resource allocation to different activities is done using which of the following representations?
 - 1 PERT chart
 - 2 activity network representation
 - 3 work breakdown structure
 - 4 Gantt chart